Application on joint species distribution model

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Scaling up from GLMs to GLMMs

Joint species distribution models are just glorified generalised linear mixed-effect models (GLMMs)

Single-species GLM

The response of species A to environment X across sites.

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Whiteboard: graphical way

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Whiteboard: expand to multi-species dataset

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Whiteboard: priors and random effects (if covered in Part 1)

Detour 1: incorporating traits

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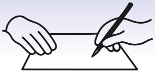
Whiteboard: link to multivariate ordination

Fitting JSDM (finaaaaally)



- Recommended packages (with traits and latent variables)
  - `boral` (Bayesian)
  - `gllvm` (Frequentist)

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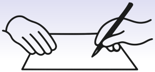

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  - Fast
  - Easy-to-learn syntax

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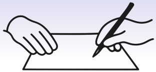

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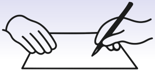
Pair up with your laptops

`https://github.com/hrlai/bayes-workshop`

Coding session



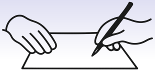
Run `code/part2.R` up until the `brm` section.

Model diagnostics

Sometimes classical statistics gives up.
Bayes never gives up ... so we're under more responsibility to check our models.

— *Andrew Gelman*

Post-processing posteriors



You get posterior for everything.

This is one of the perks of Bayesian inference.

Whiteboard: why we should calculate eveything across posteriors

Bonus: information borrowing

Parameters can be correlated;
this can be a good thing called *shrinkage*.

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- At which point you may find that you don't need Bayes anyways
- I still use frequentist approach, and everytime I return to it, I realise how much more I appreciate frequentist because of what I have learnt from Bayesian inference.

Thank you!

Feedback welcome! Link to Google Form below:

On the GitHub repository:

- Further readings
- Suggested `R` packages